Homework 3

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2026-10-05

```
library(nlme)
```

```
## Warning: package 'nlme' was built under R version 4.5.2
```

```
library(ggplot2)
```

```
## Warning: package 'ggplot2' was built under R version 4.5.2
```

```
library(dplyr)
```

```
## Warning: package 'dplyr' was built under R version 4.5.2
```

```
##
## Attaching package: 'dplyr'
```

```
## The following object is masked from 'package:nlme':
##
##     collapse
```

```
## The following objects are masked from 'package:stats':
##
##     filter, lag
```

```
## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

1. load data

```
# Read all lines and find where data starts
all_lines <- readLines("vv.txt")
```

```
## Warning in readLines("vv.txt"): incomplete final line found on 'vv.txt'
```

```r
# Find the line that starts the numeric data
data_start <- which(grepl("^[[:space:]]*[0-9]+[[:space:]]+[0-9]", all_lines))[1]
cat("Data starts at line:", data_start, "\n")
```

```
## Data starts at line: 47
```

```r
# Read the data starting from the first data line
vv <- read.table("vv.txt", skip = data_start - 1, header = FALSE)
colnames(vv) <- c("id", "smoking", "time", "FEV1")

# look at the data
head(vv, 20)
```

```
##    id smoking time FEV1
## 1   1       0    0 3.40
## 2   1       0    3 3.40
## 3   1       0    6 3.45
## 4   1       0    9 3.20
## 5   1       0   15 2.95
## 6   1       0   19 2.40
## 7   2       1    0 3.10
## 8   2       1    3 3.15
## 9   2       1    6 3.50
## 10  2       1    9 2.95
## 11  2       1   12 2.85
## 12  3       1    0 3.60
## 13  3       1    3 3.45
## 14  3       1    6 3.45
## 15  3       1    9 3.10
## 16  3       1   15 2.80
## 17  3       1   19 2.71
## 18  4       1    3 2.70
## 19  4       1    6 2.86
## 20  4       1    9 2.70
```

```r
str(vv)
```

```
## 'data.frame':    771 obs. of  4 variables:
##  $ id     : int  1 1 1 1 1 1 2 2 2 2 ...
##  $ smoking: int  0 0 0 0 0 0 1 1 1 1 ...
##  $ time   : int  0 3 6 9 15 19 0 3 6 9 ...
##  $ FEV1   : num  3.4 3.4 3.45 3.2 2.95 2.4 3.1 3.15 3.5 2.95 ...
```

```r
summary(vv)
```

```
##        id            smoking            time            FEV1      
##  Min.   :  1.00   Min.   :0.0000   Min.   : 0.00   Min.   :1.180  
##  1st Qu.: 33.50   1st Qu.:1.0000   1st Qu.: 3.00   1st Qu.:2.550  
##  Median : 66.00   Median :1.0000   Median : 9.00   Median :3.000  
##  Mean   : 66.22   Mean   :0.7549   Mean   : 8.84   Mean   :2.986  
##  3rd Qu.: 98.50   3rd Qu.:1.0000   3rd Qu.:15.00   3rd Qu.:3.400  
##  Max.   :133.00   Max.   :1.0000   Max.   :19.00   Max.   :4.900
```

```r
dim(vv)
```

```
## [1] 771   4
```

```r
#just for sanity
cat("Number of subjects:", length(unique(vv$id)), "\n")
```

```
## Number of subjects: 133
```

```r
cat("Total observations:", nrow(vv), "\n")
```

```
## Total observations: 771
```

```r
# make variables numeric and factor for easier analysis and group assignment
vv$time_numeric <- as.numeric(vv$time)
vv$smoking_label <- factor(vv$smoking,
                           levels = c(0, 1),
                           labels = c("Former Smoker", "Current Smoker"))

# Sort by subject and time
vv <- vv[order(vv$id, vv$time_numeric), ]

# Create grouped data object for nlme,
vv_grouped <- groupedData(FEV1 ~ time_numeric | id, data = vv)
head(vv_grouped)
```

```
## Grouped Data: FEV1 ~ time_numeric | id
##   id smoking time FEV1 time_numeric smoking_label
## 1  1       0    0 3.40            0 Former Smoker
## 2  1       0    3 3.40            3 Former Smoker
## 3  1       0    6 3.45            6 Former Smoker
## 4  1       0    9 3.20            9 Former Smoker
## 5  1       0   15 2.95           15 Former Smoker
## 6  1       0   19 2.40           19 Former Smoker
```

2. make a trajectory plot

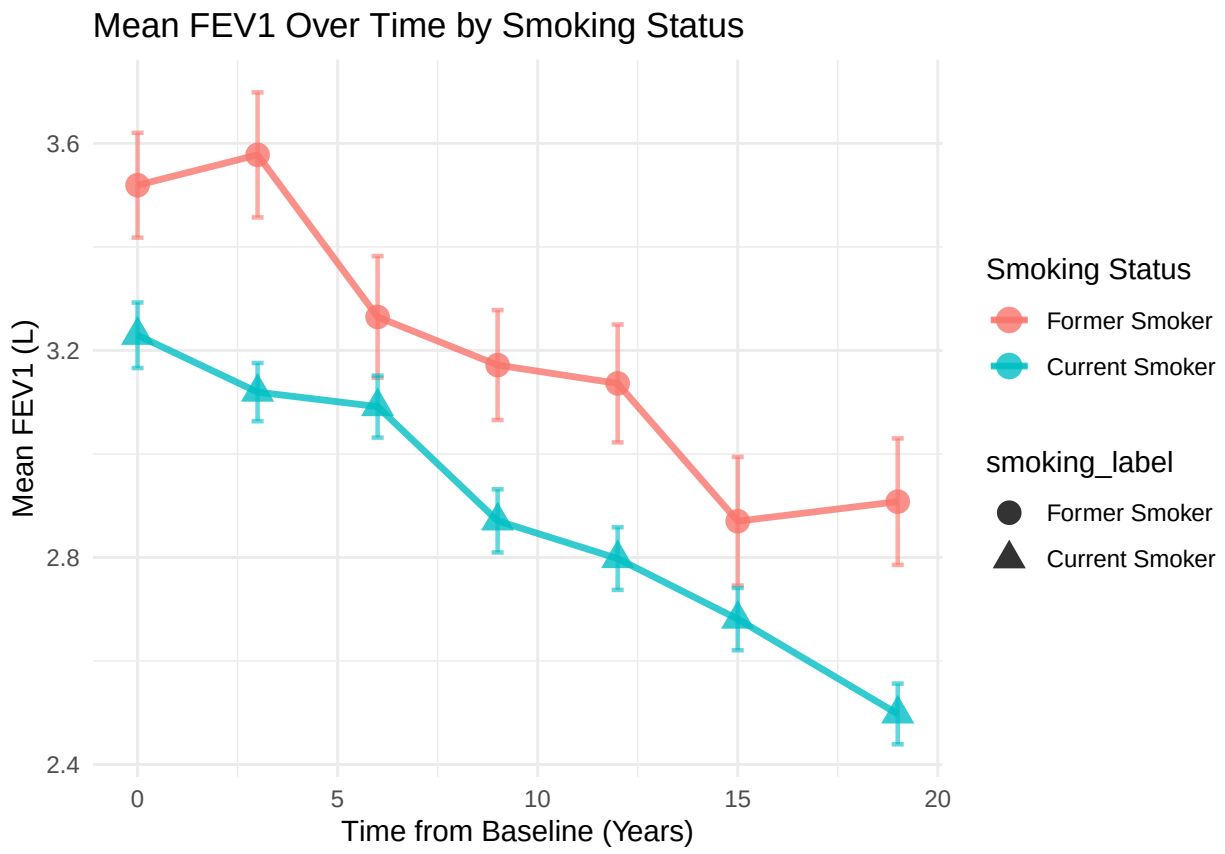
```r
#group by time and smoking status, and get mean, sd, sum, and se for FEV1
trajectory_data <- vv %>%
  group_by(time_numeric, smoking_label) %>%
  summarize(
    mean_fev1 = mean(FEV1, na.rm = TRUE),
    sd_fev1 = sd(FEV1, na.rm = TRUE),
    n_obs = sum(!is.na(FEV1)),
    se_fev1 = sd_fev1 / sqrt(n_obs),
    .groups = 'drop'
  ) %>%
  arrange(time_numeric) # in order
head(trajectory_data)
```

```
## # A tibble: 6 x 6
##   time_numeric smoking_label  mean_fev1 sd_fev1 n_obs se_fev1
##          <dbl> <fct>              <dbl>   <dbl> <int>   <dbl>
## 1            0 Former Smoker       3.52   0.486    23  0.101
## 2            0 Current Smoker      3.23   0.583    85  0.0633
## 3            3 Former Smoker       3.58   0.628    27  0.121
## 4            3 Current Smoker      3.12   0.548    95  0.0563
## 5            6 Former Smoker       3.26   0.622    28  0.118
## 6            6 Current Smoker      3.09   0.565    89  0.0599
```

```
# Plot
plot_trajectory <- ggplot(trajectory_data,
                          aes(x = time_numeric,
                              y = mean_fev1,
                              color = smoking_label,     # easy to visualize
                              shape = smoking_label,
                              group = smoking_label)) +
  geom_line(size = 1.2, alpha = 0.8) +                   # just aesthetics
  geom_point(size = 4, alpha = 0.8) +
  geom_errorbar(aes(ymin = mean_fev1 - se_fev1,
                    ymax = mean_fev1 + se_fev1),
                width = 0.3, size = 0.8, alpha = 0.6) +
  labs(
    title = "Mean FEV1 Over Time by Smoking Status",
    x = "Time from Baseline (Years)",
    y = "Mean FEV1 (L)",
    color = "Smoking Status"
  ) +
  theme_minimal()
```

```
## Warning: Using ‘size‘ aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use ‘linewidth‘ instead.
## This warning is displayed once per session.
## Call ‘lifecycle::last_lifecycle_warnings()‘ to see where this warning was
## generated.
```

```
print(plot_trajectory)
```



3. fit three models

Linear

```
model_linear <- gls(FEV1 ~ time_numeric * smoking,
                    data = vv_grouped,
                    correlation = corCompSymm(form = ~ 1 | id),
                    na.action = na.omit,
                    method = "REML") # might need changing?
print(summary(model_linear))
```

```
## Generalized least squares fit by REML
##   Model: FEV1 ~ time_numeric * smoking
##   Data: vv_grouped
##        AIC      BIC    logLik
##   323.6031 351.4581 -155.8016
##
## Correlation Structure: Compound symmetry
##  Formula: ~1 | id
##  Parameter estimate(s):
##       Rho
## 0.8595179
```

```
##
## Coefficients:
##                           Value  Std.Error   t-value p-value
## (Intercept)            3.507677 0.09804324  35.77684  0.0000
## time_numeric          -0.033852 0.00262840 -12.87912  0.0000
## smoking               -0.272676 0.11239910  -2.42596  0.0155
## time_numeric:smoking -0.004570 0.00301829  -1.51419  0.1304
##
##  Correlation:
##                       (Intr) tm_nmr smokng
## time_numeric          -0.250
## smoking               -0.872  0.218
## time_numeric:smoking  0.218 -0.871 -0.246
##
## Standardized residuals:
##         Min          Q1         Med          Q3         Max
## -2.97625810 -0.65102769  0.01890054  0.68738226  3.14895565
##
## Residual standard error: 0.5711548
## Degrees of freedom: 771 total; 767 residual
```

```
ll_linear <- as.numeric(logLik(model_linear)) #log likelihood
np_linear <- attr(logLik(model_linear), "df") #finding the number of parametrs

#-155.8016 and 6, respectively
```

Saturated

```
model_saturated <- gls(FEV1 ~ factor(time_numeric) * smoking, #time is a factor
                       data = vv_grouped,
                       correlation = corCompSymm(form = ~ 1 | id),
                       na.action = na.omit,
                       method = "REML")
print(summary(model_saturated))
```

```
## Generalized least squares fit by REML
##   Model: FEV1 ~ factor(time_numeric) * smoking
##   Data: vv_grouped
##        AIC      BIC    logLik
##   360.0919 434.1617 -164.0459
##
## Correlation Structure: Compound symmetry
##  Formula: ~1 | id
##  Parameter estimate(s):
##       Rho
## 0.8610401
##
## Coefficients:
##                                    Value  Std.Error  t-value p-value
## (Intercept)                     3.494945 0.10400506 33.60361  0.0000
## factor(time_numeric)3          -0.024091 0.06120601 -0.39360  0.6940
```

```
## factor(time_numeric)6          -0.224264 0.06103884 -3.67411  0.0003
## factor(time_numeric)9          -0.319301 0.06008601 -5.31407  0.0000
## factor(time_numeric)12         -0.393770 0.06042159 -6.51705  0.0000
## factor(time_numeric)15         -0.510895 0.06367754 -8.02316  0.0000
## factor(time_numeric)19         -0.605545 0.06125946 -9.88492  0.0000
## smoking                        -0.280677 0.11889689 -2.36068  0.0185
## factor(time_numeric)3:smoking  -0.070654 0.06914339 -1.02185  0.3072
## factor(time_numeric)6:smoking   0.075837 0.06925835  1.09499  0.2739
## factor(time_numeric)9:smoking  -0.059949 0.06862269 -0.87360  0.3826
## factor(time_numeric)12:smoking -0.021252 0.06916802 -0.30725  0.7587
## factor(time_numeric)15:smoking -0.064180 0.07252721 -0.88491  0.3765
## factor(time_numeric)19:smoking -0.099580 0.07038328 -1.41482  0.1575
##
##  Correlation:
##                                (Intr) fc(_)3 fc(_)6 fc(_)9 fc(_)12 fc(_)15
## factor(time_numeric)3          -0.321
## factor(time_numeric)6          -0.326  0.549
## factor(time_numeric)9          -0.332  0.557  0.566
## factor(time_numeric)12         -0.329  0.551  0.560  0.571
## factor(time_numeric)15         -0.313  0.522  0.535  0.550  0.536
## factor(time_numeric)19         -0.327  0.548  0.557  0.568  0.565   0.537
## smoking                        -0.875  0.281  0.285  0.291  0.288   0.274
## factor(time_numeric)3:smoking   0.284 -0.885 -0.486 -0.493 -0.488  -0.462
## factor(time_numeric)6:smoking   0.288 -0.484 -0.881 -0.499 -0.494  -0.471
## factor(time_numeric)9:smoking   0.291 -0.488 -0.496 -0.876 -0.500  -0.481
## factor(time_numeric)12:smoking  0.287 -0.482 -0.489 -0.499 -0.874  -0.469
## factor(time_numeric)15:smoking  0.275 -0.458 -0.469 -0.483 -0.471  -0.878
## factor(time_numeric)19:smoking  0.285 -0.477 -0.485 -0.495 -0.492  -0.467
##                                fc(_)19 smokng f(_)3: f(_)6: f(_)9: f(_)12:
## factor(time_numeric)3
## factor(time_numeric)6
## factor(time_numeric)9
## factor(time_numeric)12
## factor(time_numeric)15
## factor(time_numeric)19
## smoking                         0.286
## factor(time_numeric)3:smoking  -0.485  -0.316
## factor(time_numeric)6:smoking  -0.491  -0.318  0.544
## factor(time_numeric)9:smoking  -0.498  -0.322  0.549  0.554
## factor(time_numeric)12:smoking -0.494  -0.318  0.542  0.546  0.554
## factor(time_numeric)15:smoking -0.471  -0.304  0.516  0.523  0.533  0.521
## factor(time_numeric)19:smoking -0.870  -0.315  0.537  0.541  0.547  0.544
##                                f(_)15:
## factor(time_numeric)3
## factor(time_numeric)6
## factor(time_numeric)9
## factor(time_numeric)12
## factor(time_numeric)15
## factor(time_numeric)19
## smoking
## factor(time_numeric)3:smoking
## factor(time_numeric)6:smoking
## factor(time_numeric)9:smoking
## factor(time_numeric)12:smoking
```

```
## factor(time_numeric)15:smoking
## factor(time_numeric)19:smoking  0.520
##
## Standardized residuals:
##         Min          Q1         Med          Q3         Max
## -2.95345265 -0.64795471  0.04271473  0.67648856  3.15474240
##
## Residual standard error: 0.5701971
## Degrees of freedom: 771 total; 757 residual
```

```
ll_saturated <- as.numeric(logLik(model_saturated))
np_saturated <- attr(logLik(model_saturated), "df")

#-164.0459 and 16, respectively
```

Quadratic

```
model_quadratic <- gls(FEV1 ~ (time_numeric + I(time_numeric^2)) * smoking,
                       data = vv_grouped,
                       correlation = corCompSymm(form = ~ 1 | id),
                       na.action = na.omit,
                       method = "REML")
print(summary(model_quadratic))
```

```
## Generalized least squares fit by REML
##   Model: FEV1 ~ (time_numeric + I(time_numeric^2)) * smoking
##   Data: vv_grouped
##        AIC      BIC    logLik
##   354.8476 391.9666 -169.4238
##
## Correlation Structure: Compound symmetry
##  Formula: ~1 | id
##  Parameter estimate(s):
##       Rho
## 0.8592111
##
## Coefficients:
##                               Value  Std.Error  t-value p-value
## (Intercept)                3.528792 0.10123832 34.85628  0.0000
## time_numeric              -0.041108 0.00909908 -4.51781  0.0000
## I(time_numeric^2)          0.000376 0.00045146  0.83302  0.4051
## smoking                   -0.300615 0.11588091 -2.59418  0.0097
## time_numeric:smoking       0.005310 0.01042199  0.50953  0.6105
## I(time_numeric^2):smoking -0.000517 0.00052169 -0.99146  0.3218
##
##  Correlation:
##                           (Intr) tm_nmr I(t_^2) smokng tm_nm:
## time_numeric              -0.310
## I(time_numeric^2)          0.250 -0.957
## smoking                   -0.874  0.271 -0.219
## time_numeric:smoking       0.270 -0.873  0.836  -0.303
```

```
## I(time_numeric^2):smoking -0.217  0.828 -0.865   0.244 -0.957
##
## Standardized residuals:
##         Min          Q1         Med          Q3         Max
## -2.97145338 -0.66289760  0.01876922  0.68446199  3.17032064
##
## Residual standard error: 0.5710302
## Degrees of freedom: 771 total; 765 residual
```

```
ll_quadratic <- as.numeric(logLik(model_quadratic))
np_quadratic <- attr(logLik(model_quadratic), "df")

#-169.4238 and 8, respectively
```

All of the models are compared to each other and have the outputs of that comparison formatted in the same way for eventual knitting to .pdf

```
# LIKELIHOOD RATIO TESTS and ALPHA = 0.01

# Linear vs Saturated
lrt_1 <- -2 * (ll_linear - ll_saturated)
df_1 <- np_saturated - np_linear
p_1 <- 1 - pchisq(lrt_1, df = df_1)

#this is just for printing out the results of lrt, df, and p for this test
cat("LRT Statistic:", round(lrt_1, 4), "\n")
```

```
## LRT Statistic: -16.4887
```

```
cat("DF:", df_1, "\n")
```

```
## DF: 10
```

```
cat("P-value:", round(p_1, 6), "\n")
```

```
## P-value: 1
```

```
#if else loops because why not
if (p_1 < 0.01) {
  cat("Decision: Linear INADEQUATE\n")
  test1_adequate <- FALSE
} else {
  cat("Decision: Linear ADEQUATE\n")
  test1_adequate <- TRUE
}
```

```
## Decision: Linear ADEQUATE
```

```
# Quadratic vs Saturated
cat("\n\nTEST 2: QUADRATIC vs SATURATED\n")
```

```
##
##
## TEST 2: QUADRATIC vs SATURATED
```

```
lrt_2 <- -2 * (ll_quadratic - ll_saturated)
df_2 <- np_saturated - np_quadratic
p_2 <- 1 - pchisq(lrt_2, df = df_2)

cat("LRT Statistic:", round(lrt_2, 4), "\n")
```

```
## LRT Statistic: 10.7558
```

```
cat("DF:", df_2, "\n")
```

```
## DF: 8
```

```
cat("P-value:", round(p_2, 6), "\n")
```

```
## P-value: 0.215925
```

```
if (p_2 < 0.01) {
  cat("Decision: Quadratic INADEQUATE\n")
  test2_adequate <- FALSE
} else {
  cat("Decision: Quadratic ADEQUATE\n")
  test2_adequate <- TRUE
}
```

```
## Decision: Quadratic ADEQUATE
```

```
# TEST 3: Linear vs Quadratic
cat("\n\nTEST 3: LINEAR vs QUADRATIC\n")
```

```
##
##
## TEST 3: LINEAR vs QUADRATIC
```

```
lrt_3 <- -2 * (ll_linear - ll_quadratic)
df_3 <- np_quadratic - np_linear
p_3 <- 1 - pchisq(lrt_3, df = df_3)

cat("LRT Statistic:", round(lrt_3, 4), "\n")
```

```
## LRT Statistic: -27.2445
```

```
cat("DF:", df_3, "\n")
```

```
## DF: 2
```

```r
cat("P-value:", round(p_3, 6), "\n")
```

```
## P-value: 1
```

```r
if (p_3 < 0.01) {
  cat("Decision: Linear INADEQUATE vs Quadratic\n")
} else {
  cat("Decision: Linear ADEQUATE vs Quadratic\n")
}
```

```
## Decision: Linear ADEQUATE vs Quadratic
```

```r
#model selection based on adequacy in previous chunk

if (test1_adequate) {
  selected_model <- model_linear
  model_name <- "LINEAR"
  cat("Selected: LINEAR MODEL\n")
} else if (test2_adequate) {
  selected_model <- model_quadratic
  model_name <- "QUADRATIC"
  cat("Selected: QUADRATIC MODEL\n")
} else {
  selected_model <- model_saturated
  model_name <- "SATURATED"
  cat("Selected: SATURATED MODEL\n")
}
```

```
## Selected: LINEAR MODEL
```

```r
cat("Model for predictions:", model_name, "\n\n")
```

```
## Model for predictions: LINEAR
```

The linear is the chosen model here!

4. FEV1 change predictions

```r
# create prediction dataset
pred_data <- expand.grid(
  time_numeric = c(3, 6, 9, 15),
  smoking = c(0, 1)
)
pred_data$smoking_label <- factor(pred_data$smoking,
                                  levels = c(0, 1),
                                  labels = c("Former Smoker", "Current Smoker"))

# Get predictions
```

```
pred_data$predicted_fev1 <- predict(selected_model,
                                    newdata = pred_data,
                                    level = 0)
head(pred_data)
```

```
##   time_numeric smoking  smoking_label predicted_fev1
## 1            3       0  Former Smoker       3.406123
## 2            6       0  Former Smoker       3.304568
## 3            9       0  Former Smoker       3.203013
## 4           15       0  Former Smoker       2.999904
## 5            3       1 Current Smoker       3.119736
## 6            6       1 Current Smoker       3.004470
```

Dividng up by year 3 to 6, 9 to 15, never smokers, and between groups

```
# year 3 to 6

fev1_3_current <- pred_data$predicted_fev1[pred_data$time_numeric == 3 & pred_data$smoking == 1]
fev1_6_current <- pred_data$predicted_fev1[pred_data$time_numeric == 6 &  pred_data$smoking == 1]
change_3_6_current <- fev1_6_current - fev1_3_current

fev1_3_former <- pred_data$predicted_fev1[pred_data$time_numeric == 3 & pred_data$smoking == 0]
fev1_6_former <- pred_data$predicted_fev1[pred_data$time_numeric == 6 & pred_data$smoking == 0]
change_3_6_former <- fev1_6_former - fev1_3_former

cat("Year 3 to 6, Always-Smokers:  Change =", round(change_3_6_current, 4), "L\n")
```

```
## Year 3 to 6, Always-Smokers:  Change = -0.1153 L
```

```
cat("Year 3 to 6, Never-Smokers:   Change =", round(change_3_6_former, 4), "L\n")
```

```
## Year 3 to 6, Never-Smokers:   Change = -0.1016 L
```

```
# year 9 to 15
fev1_9_current <- pred_data$predicted_fev1[pred_data$time_numeric == 9 & pred_data$smoking == 1]
fev1_15_current <- pred_data$predicted_fev1[pred_data$time_numeric == 15 & pred_data$smoking == 1]
change_9_15_current <- fev1_15_current - fev1_9_current

fev1_9_former <- pred_data$predicted_fev1[pred_data$time_numeric == 9 & pred_data$smoking == 0]
fev1_15_former <- pred_data$predicted_fev1[pred_data$time_numeric == 15 & pred_data$smoking == 0]
change_9_15_former <- fev1_15_former - fev1_9_former

cat("Year 9 to 15, Always-Smokers:  Change =", round(change_9_15_current, 4), "L\n")
```

```
## Year 9 to 15, Always-Smokers:  Change = -0.2305 L
```

```
cat("Year 9 to 15, Never-Smokers:   Change =", round(change_9_15_former, 4), "L\n")
```

```
## Year 9 to 15, Never-Smokers:   Change = -0.2031 L
```